

Reviewer Pack — Quandle Isomorphism Complexity and ACC^0 Circuit Depth

Entry c39cd0b7c42e · INCONCLUSIVE

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Quandle Isomorphism Complexity and ACC^0 Circuit Depth

Entry ID: c39cd0b7c42e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-08 19:22:50 UTC

1. Conjecture

Field A (mathematical branch): Quandle Theory **Field B** (complexity object): ACC^0 Circuit Complexity

Statement:

For any finite quandle Q of size n , the minimum depth of an ACC^0 circuit computing the quandle isomorphism problem is $\Omega(\log n)$. Equivalently, the quandle isomorphism problem cannot be solved by constant-depth ACC^0 circuits for any $n \geq 2$.

Rationale (proposer's reasoning):

Quandles, algebraic structures encoding knot-theoretic symmetries, exhibit complex combinatorial properties. Their isomorphism problem, though seemingly abstract, may mirror the inherent difficulty of circuit lower bounds. By analyzing the algebraic constraints of quandles, we hypothesize that their structural richness necessitates super-constant depth for efficient computation.

Taxonomy category: ACC^0 Lower Bounds (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 78deef625403b109

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.95	SAFE	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.95	HITS	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def generate_quandle(n):
    elements = list(range(1, n + 1))
    operation_table = {}
    for i in range(n):
        for j in range(n):
            operation_table[(i, j)] = (i + j) % n + 1
    return operation_table

def are_isomorphic(q1, q2):
    if len(q1) != len(q2):
        return False
    n = len(q1)
    for perm in itertools.permutations(range(1, n + 1)):
        is_isomorphic = True
        for i in range(n):
            for j in range(n):
                if q1[(i, j)] != q2[perm[i] - 1][perm[j] - 1]:
                    is_isomorphic = False
                    break
            if not is_isomorphic:
                break
        if is_isomorphic:
            return True
    return False

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    q1 = generate_quandle(n)
    q2 = generate_quandle(n)
    while are_isomorphic(q1, q2):
        q2 = generate_quandle(n)

instances_tested = 30
conjecture_holds = True
counterexample = ""
```

```

for _ in range(instances_tested):
    q3 = generate_quandle(n)
    if are_isomorphic(q1, q3) or are_isomorphic(q2, q3):
        conjecture_holds = False
        counterexample = f"Quandles {q1} and {q2} are isomorphic to {q3}"
        break

return {
    "metric_name": "Minimum Circuit Depth",
    "metric_value": math.log(n),
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, **result}}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(r["counterexample"] for r in results):
        first_failing_seed = next(s for s, r in enumerate(seeds) if r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[first_failing_seed]['counterexample']}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d648f52d.py", line 76, in <module>

```

    result = run_trial(seed)
    ~~~~~

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d648f52d.py", line 48, in run_trial

```

    while are_isomorphic(q1, q2):
    ~~~~~

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d648f52d.py", line 34, in are_isomorphic

```

    if q1[(i, j)] != q2[perm[i] - 1][perm[j] - 1]:
    ~~~~~

```

KeyError: 0

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed with KeyError before producing results, preventing evaluation of conjecture
| next: Debug test code's permutation handling and re-run with validated inputs

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	110881
2	propose	ollama_remote	qwen3:8b	0	0	120575
3	propose	ollama_remote	qwen3:8b	0	0	113051
4	propose	ollama_remote	qwen3:8b	0	0	140553
5	preregistration	ollama_remote	qwen3:8b	0	0	24092
6	novelty	ollama_remote	qwen3:8b	0	0	21483
7	novelty	ollama_remote	qwen3:8b	0	0	16015
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12551
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8169
10	judge	ollama_remote	qwen3:8b	0	0	15668

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 583036 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/c39cd0b7c42e.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/c39cd0b7c42e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/c39cd0b7c42e.tar.gz` (if generated)